

Roll No.....

ODD MID Term Examination-September (2025)

Btech Electrical 3rd Semester
Electrical Engineering Material (TEE 305)

Time: 1:30 Hrs

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing *any one of the sub questions*.
- (ii) Each question carries 10 marks

Q1	(10 marks)	CO-01
(a)	Explain different types of crystal structures with suitable examples.	
(b)	Derive Bragg's law and explain its application in X-ray diffraction.	
Q.2	(10 marks)	CO-02
(a)	Discuss the physical properties and applications of metals materials.	
(b)	Differentiate between electrical conducting and insulating materials based on their properties.	
Q3	(10 marks)	CO-05
(a)	Explain the origin of permanent magnetic dipoles in matter.	
(b)	Compare Diamagnetism, Paramagnetism, and Ferromagnetism with examples.	
Q4	(10 marks)	CO1
(a)	What are Miller Indices? Illustrate with an example.	
(b)	Discuss structural imperfections in crystals and their effects on material properties.	
Q5	(10 marks)	CO5
(a)	Explain the difference between soft and hard magnetic materials with applications.	
(b)	Write short notes on magnetostriction and its significance.	

